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Research article

Multi-assets Asian rainbow options pricing with stochastic interest rates obeying the Vasicek model

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Abstract: Asian rainbow options provide investors with a new option solution as an effective tool for asset allocation and risk management. In this paper, we address the pricing problem of Asian rainbow options with stochastic interest rates that obey the Vasicek model. By introducing the Vasicek model as the change process of the stochastic interest rate, based on the non-arbitrage principle and the stochastic differential equation, the number of assets of the Asian rainbow option is expanded to n dimensions, and the pricing formulas of the Asian rainbow option with multiple (n) assets under the Vasicek interest rate model are obtained. The multi-asset pricing results under stochastic interest rates provide more possibilities for Asian rainbow options. Furthermore, Monte Carlo simulation experiments show that the pricing formula is accurate and efficient under double stochastic errors. Finally, we perform parameter sensitivity analysis to further justify the pricing model.

Keywords: Vasicek model; rainbow options; Asian options; Monte Carlo simulation; multi-asset option
Mathematics Subject Classification: 91B70, 91G30, 91G60

1. Introduction

Asian options are path-dependent options, meaning that the payoffs are based on the average price of the underlying asset over a specific period of time [1]. Asian options can lessen their sensitivity to asset price volatility thanks to this averaging feature, which also lowers trading risk [2]. As a result, it is frequently utilized for risk management. Multi-asset options, such as rainbow options, can be used in multi-asset allocation to lower the risks posed by multiple assets. In incentive contract design and risk management, Asian rainbow options, a hybrid of Asian and rainbow options, are frequently used [3]. Asian rainbow options combine the characteristics of Asian and rainbow options. They are a multi-asset, path-dependent option. Therefore, the maximum or minimum average price of the underlying